

DATA SET
DS1509

9. Refer to Exercise 4 (Section 15.2). The data in this table are from a paired-difference experiment with $n = 7$ pairs of observations.

Population	Pair						
	1	2	3	4	5	6	7
1	8.9	8.1	9.3	7.7	10.4	8.3	7.4
2	8.8	7.4	9.0	7.8	9.9	8.1	6.9

- a. Use Wilcoxon's signed-rank test to determine whether there is a significant difference between the two populations.
b. Compare the results of part a with the result you got in Exercise 4 (Section 15.2). Are they the same? Explain.

Paired
 $d = P_1 - P_2$
 $H_0 = M_d = 0$
 $H_a: M_d \neq 0$) 對於 M_d 都是母體
的中位數
配對差值

a. Rank

$+1.5, +1, +1, -1.5, +1.5, +1, +5.5$

$$T^+ = 1.5 + 1 + 1 + 1.5 + 1.5 + 1 + 5.5 = 26.5$$

$$T^- = 1.5$$

$$\Rightarrow T = 1.5$$

$$T_{0.05, 7} = 2 > 1.5 \Rightarrow \text{reject } H_0$$

$\therefore$ 有 difference $p \rightarrow P(X > 1.5)$
約 0.146

b.

$$p = 2P(X \leq 1) = 2\left(C_1^7 \left(\frac{1}{2}\right)^7 + C_0^7 \left(\frac{1}{2}\right)^7\right) = 0.125 > 0.05$$

sign test: $6, 1-$

→ 不顯著, 不小, 接受 H_0

non reject H_0

DATA SET
DS1511

11. Machine Breakdowns The number of machine breakdowns per month was recorded for 9 months on two identical machines, A and B, used to make wire rope:

Month	A	B	d	d	Rank
1	3	7	-4	4	-6
2	14	12	2	2	+3.5
3	7	9	-2	2	-3.5
4	10	15	-5	5	-7
5	9	12	-3	3	-5
6	6	6	0	0	
7	13	12	1	1	+1.5
8	6	5	1	1	+1.5
9	7	13	-6	6	-9

- a. Do the data provide sufficient evidence to indicate a difference in the monthly breakdown rates for the two machines? Test by using a value of α near .05.
b. Can you think of a reason the breakdown rates for the two machines might vary from month to month?

$$d = A - B$$

$$H_0 = M_d = 0$$

$$H_a: M_d \neq 0$$

x No 0

有效 $n = 8$

$$T^+ = 3.5 + 1.5 + 1.5 = 6.5$$

$$T^- = 6 + 3.5 + 7 + 1.5 + 9 = 29.5$$

$$T = \min(T^+, T^-) = 6.5$$

$$T_{9, 0.05} = 4 < 6.5 \Rightarrow \text{non reject}$$

$\therefore$ no sufficient evidence of difference

b. 受到影響, 原料改變, 操作人員... 機器使用時間... 影響

15. Images and Word Recall A psychology class performed an experiment to determine whether a recall score in which instructions to form images of 25 words were given differs from an initial recall score for which no imagery instructions were given. Twenty students participated in the experiment with the results listed in the table.

Student	With Imagery	Without Imagery	Student	With Imagery	Without Imagery
1	20	5 +15	11	17	8 +9
2	24	9 +15	12	20	16 +4
3	20	5 +15	13	20	10 +10
4	18	9 +9	14	16	12 +4
5	22	6 +16	15	24	7 +17
6	19	11 +8	16	22	9 +13
7	20	8 +12	17	25	21 +4
8	19	11 +8	18	21	14 +7
9	17	7 +10	19	19	12 +7
10	21	9 +12	20	23	13 +10

- What three testing procedures can be used to test for differences in the distribution of recall scores with and without imagery? What assumptions are required for the parametric procedure? Do these data satisfy these assumptions?
- Use both the sign test and the Wilcoxon signed-rank test to test for differences in the distributions of recall scores under these two conditions.
- Compare the results of the tests in part b. Are the conclusions the same? If not, why not?

$d = \text{with} - \text{without}$

- (a) paired $\rightarrow$
1. Paired-difference t-test
 2. Sign test
 3. Wilcoxon signed-rank test

配对

每位学生测试结果必须互相配对

② 不同 student 之间: 互相独立

③ 配对差值 d 的母体 $\sim$ 常態分布

(b) Sign: $H_0: M_d = 0$
 $H_1: M_d \neq 0$

20+
0- $\rightarrow X \sim \text{Bin}(20, 0.5)$

观察 $p = 2 \left(C_{20}^{10} \left(\frac{1}{2} \right)^{20} + 0 \right)$
 $= 1.00001 < 0.05$

reject H_0

Wilcoxon Signed Rank test

$T = 0$ $T^+ = 1+2+\dots+20 = \frac{20(21)}{2} = 210$

$\Rightarrow p = 2 P(X=20) = 2 \left(\frac{1}{2} \right)^{20}$

$= 0.00001 < 0.05$

reject H_0

有 evidence 为 0 假
合逻辑

(c)
都 reject H_0

$\therefore$ imagery instruction
 $\Rightarrow$ 得分提高

6. Legos The time required for kindergarten children to assemble a specific Lego creation was measured for children who had been instructed for four different lengths of time. Five children were randomly assigned to each instructional group. The length of time (in minutes) to assemble the Lego creation was recorded for each child in the experiment.

Training Period (hours)	0.5	1.0	1.5	2.0
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8	9	4	4
14	7	6	7
9	5	7	5
12	8	6	5
10	9	6	4

Use the rank-sum (Mann-Whitney) U-test to determine whether there is a difference in the distribution of times for the four different lengths of instructional time. Use $\alpha = .01$.

to the direction
 $C = \frac{5(12+1)}{20 \times 20} = 1.5125$

H_0 : 四組裝的時間 distribution 同

H_1 : 至少一組不合

$N = 20$

$H = \frac{12}{N(N+1)} \sum \frac{R_i^2}{n_i} - 3(N+1)$

$= \frac{12}{20(21)} \left(\frac{1^2}{5} + \frac{6^2}{5} + \frac{4^2}{5} + \frac{2^2}{5} \right) - 3(21)$

$= 12.2676$

$\Rightarrow H_0 = \frac{H}{C} = \frac{12.26}{1.51} = 8.12$

$df = 4 - 1 = 3$

$\chi_{0.01, 3}^2 = 11.345$

$H_0 = 12.2676 > 11.345$

reject H_0

$\therefore$ distribution are not identical

8. Heart Rate and Exercise In Exercise 18 (Section 11.3), we presented data on the heart rates for samples of 10 men randomly selected from each of four age groups. Each man walked a treadmill at a fixed grade for a period of 12 minutes, and the increase in heart rate (the difference before and after exercise) was recorded (in beats per minute). The data are shown in the table.

	10-19	20-39	40-59	60-69
	29 (21)	24 (8)	37 (39)	28 (8)
	33 (29.5)	27 (11)	25 (10.5)	29 (11)
	26 (12.5)	33 (29.5)	22 (5.5)	34 (34)
	27 (15)	31 (24)	33 (29.5)	36 (32.5)
	39 (40)	21 (3)	28 (10)	21 (3)
	35 (36)	28 (10)	26 (12.5)	20 (1)
	33 (29.5)	24 (8)	30 (23)	25 (10.5)
	29 (21)	34 (34)	34 (34)	24 (8)
	36 (32.5)	21 (3)	27 (15)	33 (29.5)
	22 (5.5)	32 (25.5)	33 (29.5)	32 (25.5)
Total	309 (247.5)	275 (108)	295 (164)	282 (107)

- a. Do the data present sufficient evidence to indicate differences in location for at least two of the four age groups? Test using the Kruskal-Wallis H -test with $\alpha = .01$.
- b. Find the approximate p -value for the test in part a.
- c. Since the F -test in Chapter 11 and the H -test are both tests to detect differences in location of the four heart-rate populations, how do the test results compare? If the ANOVA F -test produced the test statistic $F = 0.87$ with p -value $= 0.468$, compare this p -value with the p -value in part b and explain the implications of the comparison.

H_0 : 4个 age 的 heart rate 增量 distribution 相同
 H_1 : 至少 2 个不同

$$\Rightarrow \sum = 247.5 + 164 + 216.5 + 188 = 820$$

$$a. \text{Kruskal-Wallis } H \text{ test} = \frac{12}{N(N+1)} \sum \frac{R_i^2}{n_i} - 3(N+1) = \frac{12}{40(41)} \sum \frac{R_i^2}{n_i} - 3(41) = 8.20$$

$$N = 40$$

$$H = \frac{12}{N(N+1)} \sum \frac{R_i^2}{n_i} - 3(N+1) = \frac{12}{40(41)} \sum \frac{R_i^2}{n_i} - 3(41) = 2.6310$$

$$df = k - 1 = 4 - 1 = 3$$

$$\chi_{0.01,3}^2 = 11.345 > 2.6310$$

non reject H_0

$$b. P(\chi_3^2 \geq 2.6310)$$

$$\approx 0.444 > 0.01$$

$\therefore$ non reject H_0

$$(c) F = 0.87$$

$$p_{ANOVA} = 0.468$$

> 0.1

$$p_{KW} \approx 0.444$$

$\therefore$ non reject H_0

$\therefore$ no sufficient evidence for difference